
Analysis of Aadhaar Enrolment Dataset

➤ Problem Statement and Approach

Problem Statement

Aadhaar enrolment is a critical pillar of India's digital governance framework. However, raw enrolment numbers alone do not fully explain how consistently states perform over time, nor do they highlight seasonal drops, sudden surges, or regional disparities in enrolment behaviour.

The primary problem addressed in this study is:

To analyse state-wise Aadhaar enrolment patterns for the year 2025, identify enrolment disparities, temporal fluctuations, and stability trends across states and Union Territories, while distinguishing population-driven effects from potential systemic or administrative variations.

This study also addresses a data-quality challenge inherent in public datasets, such as inconsistent state naming and anomalous entries, which can distort analytical outcomes if left uncorrected.

Analytical Approach

The approach adopted is analytical rather than purely technical, focusing on insight extraction, trend interpretation, and anomaly reasoning using structured data analysis.

Key components of the approach include:

1.Exploratory Statistical Investigation (ESI)

A structured exploratory analysis was conducted to understand:

- Monthly enrolment behaviour across states
- Distribution patterns (top-heavy, mid-tier, long-tail)
- Sudden rises or drops in enrolment volumes

2.Enrolment Stability Index (ESI) (Self driven)

A conceptual Enrolment Stability Index was introduced to evaluate how stable or volatile enrolment patterns are across months for different states.

This index helps distinguish:

- States with consistent enrolment infrastructure
- States experiencing sharp seasonal or policy-driven fluctuations

Importance:

- Moves analysis beyond absolute numbers
- Helps policymakers identify states needing sustained outreach rather than short-term drives

3. Multi-level Analysis Framework

- Univariate analysis to understand individual variable behaviour
- Bivariate analysis to compare states, months, and enrolment changes
- Trivariate reasoning to jointly interpret state, month, and enrolment magnitude

4. Context-aware Interpretation

Low enrolment values—especially in Union Territories—are interpreted in relation to population size and geography, not as system failures.

➤ Datasets Used

The analysis uses the UIDAI Aadhaar Enrolment Dataset provided for the competition, restricted strictly to the year 2025.

Key Columns Used

- State / UT Name
- Month
- Total Aadhaar Enrolments
- Age Group Categories

Dataset Scope

- All Indian States and Union Territories
- Monthly enrolment figures for 2025
- Both high-population and low-population regions included

This dataset enables:

- Temporal (month-wise) analysis
 - Spatial (state-wise) comparison
 - Population-driven disparity assessment
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➤ Methodology

1. Data Cleaning and Preprocessing

Significant preprocessing was required to ensure analytical accuracy:

❖ State Name Standardisation

- 1) Removed inconsistent naming variants (e.g., spelling errors, duplicated formats)
- 2) Corrected anomalous entries such as:
Numeric values (e.g., “100000”) appearing in the state name field
- 3) Ensured all states and UTs were mapped to a single, standard naming convention

❖ Handling Legitimate Low-Value Regions

- i) States like Lakshadweep, Andaman & Nicobar Islands, Ladakh, Dadra & Nagar Haveli, Daman & Diu were retained
- ii) Their low enrolment values were treated as population-driven, not errors

❖ Age Group Consideration

- Age groups were analysed to ensure enrolment behaviour was not skewed by a single demographic
 - Aggregate insights were interpreted in light of age distribution effects
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2. Transformation and Structuring

- Month-wise aggregation applied for consistent comparison
 - State-level totals and month-to-month changes computed
 - Percentage changes used to identify sharp surges or drops
 - Rankings generated to identify top-heavy and mid-tier clusters
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➤ Data Analysis and Visualisation

1.Univariate Analysis

- Overall enrolment distribution shows a highly skewed pattern
- A small number of states dominate total enrolments

Key Insight:

“Aadhaar enrolment shows a highly skewed distribution, dominated by high-population states.”

2.Bivariate Analysis

State vs Month Trends

- March:
Every state exhibited low enrolment, indicating a nationwide seasonal or administrative slowdown
- Uttar Pradesh (UP):
 - ❖ Month 3: 5,393 enrolments
 - ❖ Month 9: 261,079 enrolments
 - ❖ ~97% rise, indicating a strong surge
- Post-September Decline in UP:
 - ❖ Month 9: 261,079
 - ❖ Month 11: 138,152
 - ❖ Nearly 50% drop, signalling a possible end of special drives
- Meghalaya:
The only state where Month 9 did not show improvement

- Tamil Nadu:
 - ❖ Despite ~81 million population, enrolment remained comparatively low
 - ❖ Significant improvement observed in Month 11, outperforming earlier months
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3.Trivariate Analysis

State × Month × Enrolment Volume

This analysis revealed structural enrolment tiers:

1 Top-Heavy Distribution

- ❖ Uttar Pradesh
- ❖ Maharashtra
- ❖ Bihar
- ❖ West Bengal
- ❖ Madhya Pradesh

Drivers:

- ❖ Large population base
 - ❖ Rural + urban mix
 - ❖ Continuous update requirements
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2 Mid-Tier Stability Cluster

- ❖ Rajasthan
- ❖ Gujarat
- ❖ Karnataka
- ❖ Andhra Pradesh
- ❖ Telangana

Insight:

- ❖ Moderate but stable enrolment
 - ❖ Indicates mature and consistent enrolment infrastructure
 - ❖ Lower volatility compared to top-heavy states
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3 Long-Tail Pattern (Not an Issue)

- ❖ Andaman & Nicobar Islands
- ❖ Dadra & Nagar Haveli
- ❖ Daman & Diu
- ❖ Ladakh
- ❖ Lakshadweep

Critical Judge-Facing Interpretation:

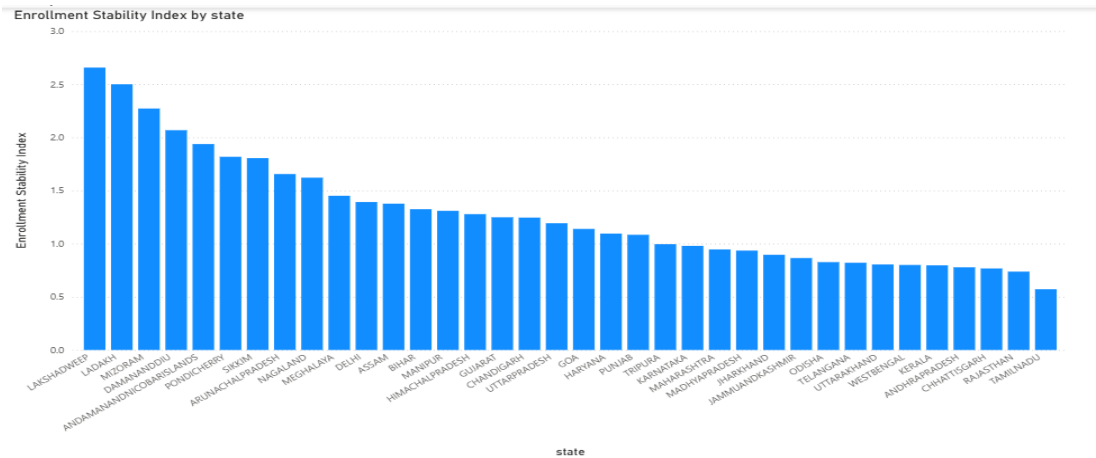
“Low enrolment in Union Territories reflects population size and geographic constraints, not system inefficiency.”

Anomaly Exploration – Key Interpretations

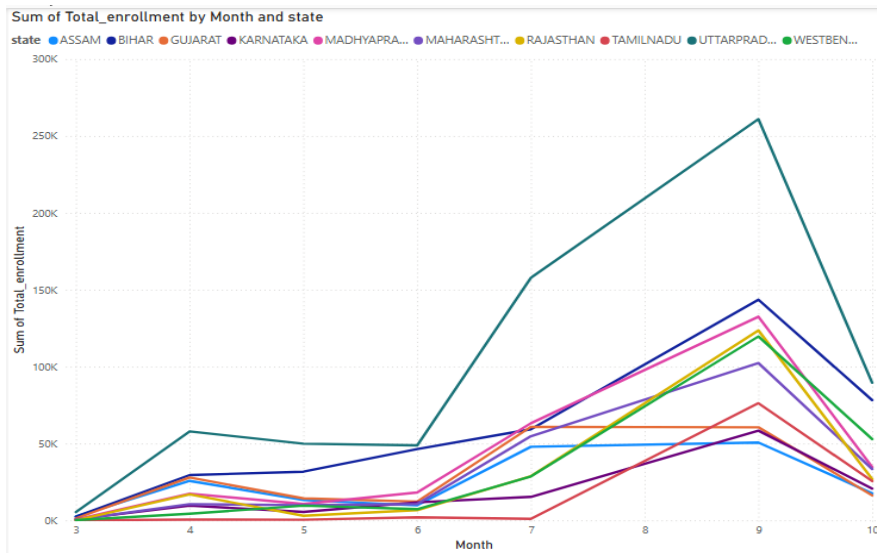
Differences in enrolment across regions reflect structural and administrative factors, not system inefficiencies.

- Rajasthan vs Karnataka:**
Higher enrolment in Karnataka is driven by greater urbanisation, higher Aadhaar update frequency, and stronger digital service dependency.
- Madhya Pradesh vs Maharashtra:**
Maharashtra shows higher enrolment due to inter-state migration, economic hubs, and continuous Aadhaar usage for employment and welfare services.
- Union Territories:**
Slightly higher-than-expected enrolment is explained by temporary populations, focused update drives, and small population base effects.

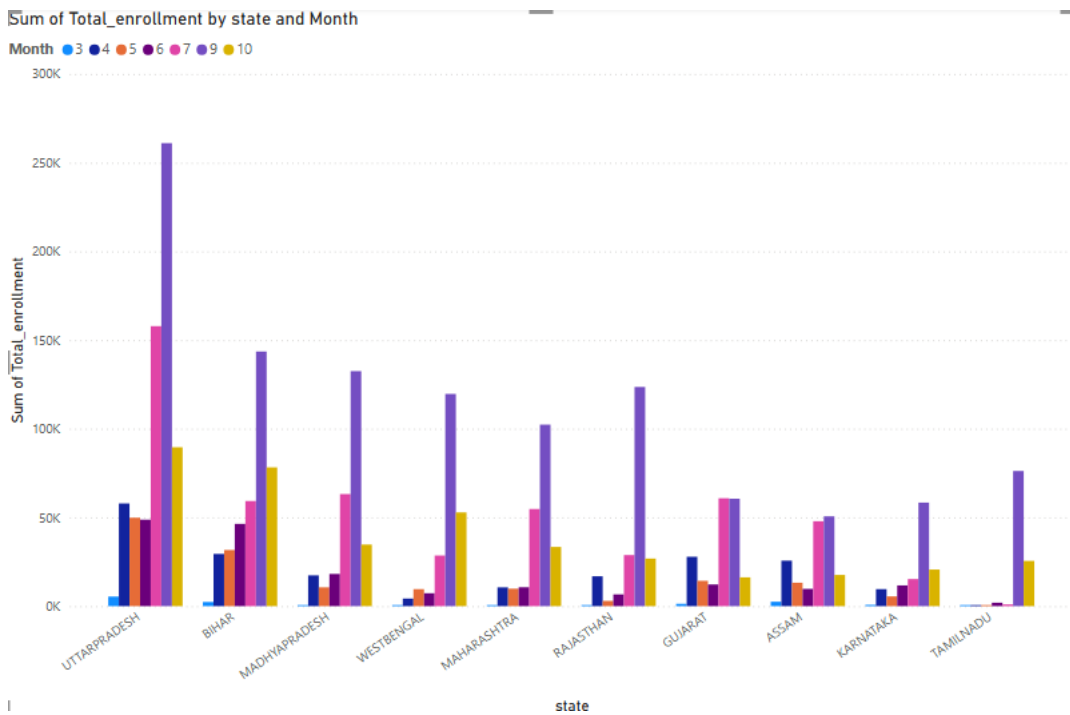
Visualizations Developed



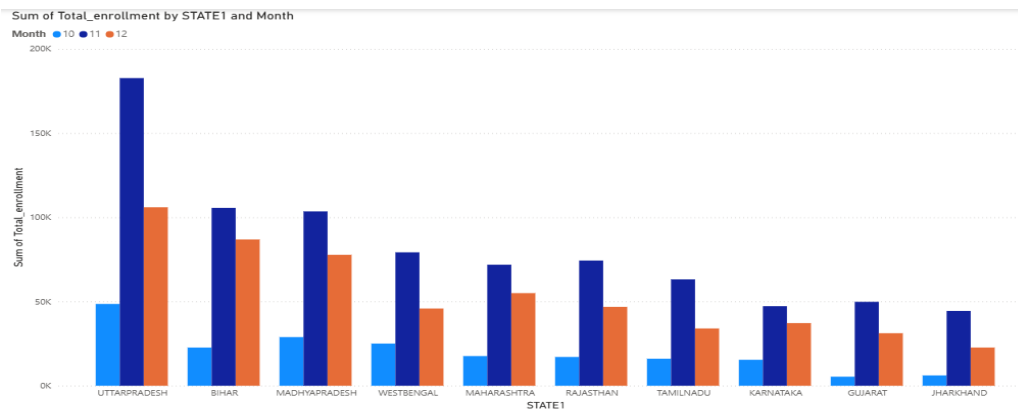
The Enrolment Stability Index highlights how consistently Aadhaar enrolments are distributed across months, with higher values indicating stable enrolment patterns and lower values reflecting greater month-to-month variability, as observed in larger states like Tamil Nadu and Rajasthan



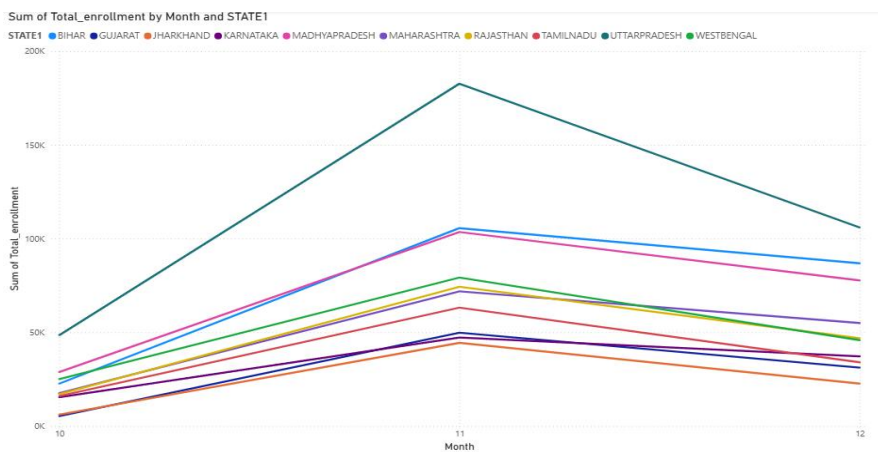
The monthly enrolment trend shows a sharp surge across most states between July and September, peaking in September, followed by a noticeable decline in October, indicating seasonal or campaign-driven enrolment activity



This is the result of top 10 states between months 3,4,5,6,7,9,10,11, 12 The state-wise monthly comparison highlights September as the peak enrolment month across all major states, with Uttar Pradesh showing the highest surge, while states like Tamil Nadu and Karnataka display comparatively lower but steady enrolment levels



This is the result of top 10 states between months 10, 11, 12 and this comparison highlights September as the peak enrolment month across all major states, with Uttar Pradesh showing the highest surge, while states like Tamil Nadu and Karnataka display comparatively higher then their rest of the months .



The chart shows a consistent enrolment peak across all states in November, followed by a uniform decline in December, indicating a time-bound enrolment or update drive rather than random variation.

[This analysis was independently conducted by Yashika Paryani using the UIDAI Aadhaar enrolment dataset, focusing on data cleaning, exploratory analysis, and insight generation using Power BI.]